Solutions for Homework Assignment #1

Answer to Question 1.

a. Intuitively, the idea is to enumerate the elements of B as follows. We start with an enumeration $a_0, a_1, a_2, \ldots$ of A (this exists since, by assumption, A is enumerable). We then obtain an enumeration of B in two stages as follows: First, because f is surjective, the sequence $f(a_0), f(a_1), f(a_2), \ldots$ contains every element of B, but some elements may appear more than once, so this is an enumeration with possible duplicates. We then obtain a proper enumeration of B (i.e., one without duplicates) by skipping the elements that have previously appeared, so that every element of B appears exactly once. We now give a more precise argument.

If B is finite, B is countable by definition and we are done. So, in the remainder of this proof we assume that B is infinite. By assumption A is countable so, by Theorem 1.10(e), it suffices to show that there is a bijection $g\colon A \to B$. Let $a_0, a_1, a_2, \ldots$ be an enumeration of A. We define g by induction as follows:

$$g(a_i) = \begin{cases} f(a_0), & \text{if } i = 0 \\ f(a_j), \text{ where } j = \min\{k \in \mathbb{N} : \ f(a_k) \notin \{g(a_0), g(a_1), \ldots, g(a_{i-1})\}, & \text{if } i > 0 \end{cases}$$

First we show that this function is well defined. That is, for each $a \in A$, there is a unique $b \in B$ such that $g(a) = b$. Since $a_0, a_1, a_2, \ldots$ is an enumeration of A, this is equivalent to proving that for every $i \in \mathbb{N}$, there is a unique $b \in B$ such that $g(a_i) = b$. We prove this by complete induction on i. The base case, $i = 0$, is obvious from the above definition. For the induction step, assume that $g(a_t)$ is well-defined for all t, $0 \le t < i$. We now show that $g(a_i)$ is also well defined: Since (1) $\{g(a_0), g(a_1), \ldots, g(a_{i-1})\}$ is finite, (2) B is infinite, and (3) the function f is onto, the set $\{k \in \mathbb{N}\colon f(a_k) \notin \{g(a_0), g(a_1), \ldots, g(a_{i-1})\}$ is not empty. By the well-ordering principle it has a minimum element, say j. By the above definition of g, we have that $g(a_i) = f(a_j)$. So $g(a_i)$ is defined. It remains to prove that g is bijective, i.e., it is injective and surjective.

<u>(i) g is injective:</u> Let $i, i' \in \mathbb{N}$ be distinct. Without loss of generality, assume that $i < i'$. Thus, by definition of g, $g(a_{i'}) = f(a_j)$ for some j such that $f(a_j) \notin \{g(a_0), g(a_1), \ldots, g(a_{i'-1})\}$. Therefore $g(a_{i'}) \neq g(a_i)$, as wanted.

<u>(ii) g is surjective:</u> Let b be an arbitrary element of B. Since f is surjective and $a_0, a_1, a_2, \ldots$ is an enumeration of A, there is some $j \in \mathbb{N}$ such that $f(a_j) = b$. Either there is some $k < j$ such that $g(a_k) = b$ or else, by the definition of g, $g(a_j) = b$. In either case, there is some i such that $g(a_i) = b$, as wanted.

b. Let $f\colon A \to B$ be an injective function. We must show that if A is uncountable then B is also uncountable. We prove the contrapositive, i.e., if B is countable then A is also countable.

If A is empty then it is countable, and we are done; so in the rest of the proof we assume that $A \neq \varnothing$ and we fix some element $a^* \in A$. Since f is injective, for any $b \in B$ there is at most one $a \in A$ such that $f(a) = b$. So we can define a function $g\colon B \to A$ as follows:

$$g(b) = \begin{cases} a, & \text{if } a \text{ is the unique element of } A \text{ such that } f(a) = b \\ a^*, & \text{if there is no } a \in A \text{ such that } f(a) = b. \end{cases}$$

The function g is surjective because for every $a \in A$ there is some $b \in B$, namely $b = f(a)$ such that $g(b) = a$. Then by part (a), if B is countable then A is countable, as wanted.

c. PROOF 1 (diagonalization): Suppose, for contradiction, that $\mathcal{P}$ is countable, and let $A_0, A_1, A_2, \ldots$ be an enumeration of $\mathcal{P}$. Let $A = \{2^i : 2^i \notin A_i\}$. A is a set of natural numbers that are powers of 2 that, for every $i \in \mathbb{N}$, differs from A_i on (at least) whether it contains 2^i. Therefore A is not in the sequence $A_0, A_1, A_2, \ldots$, contradicting that this is an enumeration of $\mathcal{P}$.

PROOF 2: For each $A \in \mathcal{P}$, let $b_0, b_1, b_2, \ldots$ be the infinite sequence of bits defined by

$$b_i = \begin{cases} 1, & \text{if } 2^i \in A \\ 0, & \text{otherwise.} \end{cases}$$

This defines a one-to-one correspondence between the elements of $\mathcal{P}$ and the set of infinite sequences of bits, which by Theorem 1.5 is uncountable. Therefore, by Theorem 1.10(e), $\mathcal{P}$ is also uncountable.

d. Let A be an uncountable set and B be a countable set. Suppose, for contradiction, that $A - B$ is countable. Then, by Theorem 1.10(c), $(A - B) \cup B$, which is equal to $A \cup B$, is countable. The function $f : A \to (A \cup B)$ defined by $f(a) = a$ is obviously injective. By (the contrapositive of) part (b), A is countable, a contradiction.

Answer to Question 2.

a. The set of functions from $\{0, 1\}$ to $\mathbb{N}$ is countable.

Reason: A function $f : \{0, 1\} \to \mathbb{N}$ can be encoded by the pair of natural numbers $\big(f(0), f(1)\big)$. Since the set of pairs of natural numbers is countable, by Theorem 1.10(e) so is the set of functions from $\{0, 1\}$ to $\mathbb{N}$.

b. The set $\mathcal{F}_{\text{inc}}$ of increasing functions from $\mathbb{N}$ to $\mathbb{N}$ is uncountable.

Reason: We use a diagonalization argument. Suppose, for contradiction, that $\mathcal{F}_{\text{inc}}$ is countable, and let $f_0, f_1, f_2, \ldots$ be an enumeration of it. We will define an increasing function $f : \mathbb{N} \to \mathbb{N}$ that is different from all functions that appear in this enumeration, giving us the desired contradiction. We define f by induction as follows:

$$f(i) = \begin{cases} f_0(0) + 1, & \text{if } i = 0 \\ \max\big(f(i-1), f_i(i)\big) + 1, & \text{if } i > 0 \end{cases}$$

Since $f(i) > f(i-1)$ for every $i > 0$, $f \in \mathcal{F}_{\text{inc}}$. Furthermore f is different from f_i for each $i \in \mathbb{N}$, because $f(i) \neq f_i(i)$. So, f is an increasing function that does not appear in the assumed enumeration of $\mathcal{F}_{\text{inc}}$, a contradiction.

c. The set $\mathcal{F}_{\text{dec}}$ of decreasing functions from $\mathbb{N}$ to $\mathbb{N}$ is countable.

Reason: The key observation here is that any function $f \in \mathcal{F}_{\text{dec}}$ is eventually constant. This is because it can decrease at most $f(0)$ times before reaching zero, and it cannot decrease further since the range of f is $\mathbb{N}$. Let $n_f \in \mathbb{N}$ be the smallest natural number at which $f \in \mathcal{F}_{\text{dec}}$ obtains its final minimum value; more precisely, (i) $n_f = 0$ or $f(n_f - 1) > f(n_f)$, and (ii) for all $n \geq n_f$, $f(n) = f(n_f)$. Thus, we can encode the function $f \in \mathcal{F}_{\text{dec}}$ by the (non-decreasing) finite sequence of natural numbers $f(0), f(1), \ldots, f(n_f)$; the remaining values of f are all equal to the last element of this sequence. By Theorem 1.4, the set of finite sequences of natural numbers is countable. Thus by Theorem 1.10(e), so is $\mathcal{F}_{\text{dec}}$.

d. The set $\mathcal{F}_{\text{mon}}$ of monotone functions from $\mathbb{N}$ to $\mathbb{N}$ is uncountable.

Reason: Let $F : \mathcal{F}_{\text{inc}} \to \mathcal{F}_{\text{mon}}$ be the function $F(f) = f$. This is obviously injective, so by Question 1(b) $\mathcal{F}_{\text{mon}}$ is uncountable since, as we proved in part (b), $\mathcal{F}_{\text{inc}}$ is uncountable.

e. The set $\mathcal{N}$ of non-monotone functions from $\mathbb{N}$ to $\mathbb{N}$ is uncountable.

Reason: We prove this using diagonalization. Suppose, for contradiction, that $\mathcal{N}$ is countable, and let $f_0, f_1, f_2, \ldots$ be an enumeration of the functions in it. We will define a function $f \in \mathcal{N}$ that differs from all f_i's, yielding the desired contradiction. Making f different from all f_i is standard; to ensure that it is non-monotone, we choose the first three values so that $f(0) < f(1)$, and $f(1) > f(2)$. With this in mind, we define $f : \mathbb{N} \to \mathbb{N}$ as follows:

$$f(i) = \begin{cases} f_0(0) + 1, & \text{if } i = 0 \\ \max\bigl(f_0(0) + 2, f_1(1) + 1\bigr), & \text{if } i = 1 \\ 0, & \text{if } i = 2 \text{ and } f_2(2) \neq 0 \\ 1, & \text{if } i = 2 \text{ and } f_2(2) = 0 \\ f_i(i) + 1, & \text{if } i > 2 \end{cases}$$

By construction, $f(0) < f(1)$ and $f(1) > f(2)$, so $f \in \mathcal{N}$. Furthermore, also by construction, $f(i) \neq f_i(i)$, so f does not appear in the sequence $f_0, f_1, f_2, \ldots$, contradicting that this is an enumeration of $\mathcal{N}$.